

Amortized Deduction for Binary Linear Systems: Learning Branching Guidance from Exact Conditional Marginals

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Abstract

A binary linear system (BLS) asks for $\mathbf{x} \in \{0, 1\}^n$ satisfying $A\mathbf{x} = \mathbf{b}$ with $A \in \{0, 1\}^{m \times n}$, a formulation that arises in laser-based non-destructive testing, where a binary occupancy image is reconstructed from integer projections. On market-split-style instances the linear relaxation is deliberately uninformative, so backtracking search spends most of its effort on branching decisions that a cheap heuristic cannot make well. We present a learned branching heuristic for this setting and an exact procedure for training it.

The method rests on two observations. First, conditioning a BLS on a partial assignment is *identical* to substituting the assignment out and obtaining a smaller BLS; a single network therefore predicts at the root and at every interior node without architectural change, and training states are ordinary instances of reduced size. Second, on instances small enough to enumerate, the exact conditional marginal $p_j = \Pr[x_j = 1 \mid A, \mathbf{b}]$ is computable, and it partitions the free variables of a node into those every surviving solution agrees on — where branching the other way empties the subtree and forces a backtrack — and those solutions disagree on, where either branch still reaches a solution. Only the first kind carries cost. This partition supplies both an exact training target and a criterion for which predictions matter: at the root only 6.3% of variables are of the first kind, but by depth 8 the share is 87.9%, which is where prediction accuracy begins converting into search cost.

We train a 21.8K-parameter bipartite graph network on these targets and use it for branching only, never for fixing variables, since it produces a probability rather than a proof. Against LP-probing — the sound procedure that detects the same variables, at one linear program each — the network needs

a single relaxation and one forward pass, so their cost ratio grows linearly in instance size: $1.7\times$ at $n = 25$, $7.2\times$ at $n = 60$, $17.8\times$ at $n = 150$, measured with a warm-started backend on both sides. Under a matched wall-clock budget on a warm-started search loop, replicated over three seeds, the resulting search is $2.6\text{--}3.0\times$ faster in median than LP-guided branching at 21×60 (faster on 23/30 instances in every seed, sign test $p = 0.005$) and expands $9.5\times$ fewer nodes, with both strategies solving every instance; within the full pipeline, where lattice reduction runs first, it closes the residual $3.0\times$ faster than LP-guided branching, while random branching closes none of it. Controlling the number of solutions across sizes, we find no measurable transfer penalty: a network trained only at 10×25 is statistically indistinguishable from one trained at the evaluation size. On a 100-instance 21×60 set the supervised guidance is $2.0\text{--}2.3\times$ faster than LP-guided branching with $5.5\times$ fewer nodes. Two further experiments locate what the supervised target does and does not capture. Training on states of deployment size — the regime where the network’s prediction margin lives — does *not* help: it loses competence on the small interior states where most nodes are spent, and the search gets slower. What does help is optimizing the objective the search actually pays: with the exact subtree cost of every branching decision available from the search itself, an actor-critic fine-tuning of the branching policy, initialized at the supervised rule, expands $2.3\times$ fewer nodes and is $2.0\text{--}2.2\times$ faster than the supervised policy on the same 100 instances ($p < 0.01$), $4.8\text{--}5.9\times$ faster than LP-guided branching. The fine-tuned policy branches on a different variable than the confidence rule at 95% of held-out states, typically one it is *less* sure about, which is direct evidence that the probability of being right and the cost of being wrong are different objectives.

We also report what the method does not achieve. CP-SAT solves the same instances $9\text{--}40\times$ faster at every size tested ($9\text{--}10\times$ at 21×60 after fine-tuning), and our system does not compete with it; decomposing that gap shows our branching produces $2.2\times$ *fewer* nodes but processes them $32\times$ more slowly, which places the deficit in per-node cost rather than in branching quality, though we do not show that it is recoverable. And the cost advantage over probing, while asymptotically real, is only $1.3\text{--}1.7\times$ at the depths where prediction matters most on our instances, where probing additionally offers proofs rather than predictions. The contribution we claim is therefore methodological: exact conditional posteriors are an obtainable and well-posed supervision signal for branching, in-tree states are a better training distribution than root states — a root-trained network is worse than

relaxation rounding when deployed in the tree, an in-tree-trained one is better by 3.4 points — and the resulting heuristic transfers across instance size. Code, generators, exact-marginal labels and per-instance results are released.

Keywords: Binary linear systems, Learning to branch, Amortized optimization, Exact conditional marginals, Market split, Graph neural networks

1. Introduction

1.1. Problem

Let $A \in \{0, 1\}^{m \times n}$ and $\mathbf{b} \in \mathbb{Z}^m$. A *binary linear system* (BLS) is the feasibility problem

$$\text{find } \mathbf{x} \in \{0, 1\}^n \text{ such that } A\mathbf{x} = \mathbf{b}, \quad (1)$$

whose solution set we write $\mathcal{S} = \{\mathbf{x} \in \{0, 1\}^n : A\mathbf{x} = \mathbf{b}\}$. Each row is a cardinality constraint $\sum_j a_{ij}x_j = b_i$ stating how many cells of a designated subset are occupied.

The motivating application is laser-based non-destructive testing. A specimen is scanned along m directions; each scan returns the integer count of absorbing cells along its path, and the task is to reconstruct the binary occupancy image. Two properties of that setting shape the problem formulation. Solutions need not be unique — several images can be consistent with the same measurements, and recovering *any* member of $\mathcal{S}$ solves the physical problem. And measurement noise can perturb $\mathbf{b}$ so that $\mathcal{S} = \emptyset$, making a reliable infeasibility verdict as valuable as a solution.

Deciding $\mathcal{S} \neq \emptyset$ is NP-complete in general. The instances studied here are drawn from a deliberately hard regime modelled on market-split problems [6, 2], in which the linear relaxation

$$\tilde{\mathcal{S}} = \{\mathbf{x} \in [0, 1]^n : A\mathbf{x} = \mathbf{b}\} \quad (2)$$

is a large polytope with predominantly fractional vertices. Relaxation-based bounding is therefore weak, and a complete solver must decide where to branch with little guidance from the relaxation — exactly the decision this paper learns.

1.2. Where a learned component attaches, and what to train it on

A complete solver for (1) alternates deduction (constraint propagation, relaxation-based pruning) with a *branching decision*: which free variable to split on, and which value to try first. Deduction is sound and must remain so; branching is a heuristic choice, where an error costs search time but never correctness. This asymmetry makes branching the natural place to attach a learned component, and it is where learning has been most successful in mixed-integer programming [9, 7].

The difficulty is supervision. Two standard answers are available and both are unsatisfactory here. Training against a planted solution $\mathbf{x}^*$ treats one member of $\mathcal{S}$ as ground truth, which is ill-posed when $|\mathcal{S}| > 1$: nothing in $(A, \mathbf{b})$ distinguishes the planted solution from any other, so the label is a sample from the posterior rather than the posterior. Imitating an expensive expert, as in strong-branching imitation, requires an expert whose decisions are both informative and affordable to collect; the natural candidate here, LP-probing, costs one linear program per variable per node.

We take a third route, available because these instances have planted solutions and because feasibility can be enumerated at moderate size: compute the *exact conditional posterior* by exhaustive enumeration on small instances, and train against it. Section 3 shows that this is both tractable and, in a precise sense, the right target — and that the posterior additionally identifies *which* predictions can affect search at all.

1.3. Contributions

1. **A reduction identity that makes in-tree supervision trivial to implement.** Conditioning (1) on a partial assignment yields a smaller BLS with the same structure (Proposition 1). One architecture, one label generator and one training loop therefore serve the root and every interior node, and a depth- d training state is stored as an ordinary instance on $n - d$ variables.
2. **A partition of branching decisions by whether they can cost anything.** The exact marginals split a node’s free variables into DET (all surviving solutions agree; branching the other way empties the subtree) and OPEN (solutions disagree; either branch reaches a solution). Only DET errors cost a backtrack. We measure the composition: 6.3% DET at the root, rising to 87.9% at depth 8 (Sections 4.3, 4.4). This locates useful prediction inside the tree rather than at the root and explains why root-level accuracy is a poor objective.

- 60 3. **A cost characterization of the learned heuristic against the**
61 **sound procedure it approximates.** The network needs one relax-
62 ation and one forward pass; LP-probing needs one relaxation per vari-
63 able. Measured with a warm-started backend on both sides, the ratio
64 scales from $1.7\times$ at $n = 25$ to $17.8\times$ at $n = 150$ (Section 4.5). We
65 report this as a scaling law rather than a single figure, and note that
66 at the sizes where our search operates the advantage is modest and
67 probing has the additional merit of soundness.
- 68 4. **End-to-end evaluation under matched wall-clock budgets.** At
69 21×60 on a warm-started search loop, replicated over three seeds, the
70 learned guidance is $2.0\text{--}2.3\times$ faster in median than LP-based branching
71 on a 100-instance set (faster on 70–71/100, sign test $p < 0.001$) and
72 expands $5.5\times$ fewer nodes; on the original 30-instance set the margins
73 are $2.6\text{--}3.0\times$ and $9.5\times$, and both strategies solve every instance within
74 600 s (Section 4.6). In the deployed cascade both sound strategies close
75 every residual instance, the learned one $3.0\times$ faster (40/50 wins pooled
76 over seeds, $p = 2 \times 10^{-5}$), while random branching closes none of the
77 50 residual instances (Section 4.12).
- 78 5. **Size transfer with solution multiplicity controlled.** Holding $|\mathcal{S}|$
79 comparable across sizes, a network trained only at 10×25 matches one
80 trained at the evaluation size 21×60 (faster on 15/30 instances; total
81 time differing by 0.4%), showing the learned quantity is not size-specific
82 (Section 4.8).
- 83 6. **A negative result on aligning the training distribution with**
84 **deployment.** The network’s margin over relaxation rounding lives on
85 states with 50–60 free variables, outside its 13–25 training range. Train-
86 ing on exactly those states, with everything else fixed, raises accuracy
87 there by about one point and *doubles* the search tree, because it loses
88 competence on the small interior states where two thirds of all nodes
89 are spent (Section 4.10).
- 90 7. **Cost-aware fine-tuning of the branching policy.** The exact cost
91 of a branching decision — the number of nodes its subtree expands —
92 is available from the search itself, and on held-out states the confidence
93 rule picks the cheapest of its five candidates only 22% of the time.
94 An actor-critic fine-tuning on that cost, initialized at the supervised
95 rule with the marginal network frozen, expands $2.3\times$ fewer nodes and
96 is $2.0\text{--}2.2\times$ faster than the supervised policy on the 100-instance set
97 (70/100 and 64–67/100 wins, $p < 0.01$ in every seed), $4.8\text{--}5.9\times$ faster

98 than LP-guided branching (Section 4.11).

99 2. Related Work

100 2.1. Binary linear systems and market-split hardness

101 Problem (1) is a 0/1 integer feasibility problem and is NP-complete for
102 general A . Practical hardness, however, depends on the instance distribution.
103 Cornuéjols and Dawande [6] introduced the market-split family, in which m
104 agents must divide n items so that every agent’s budget is met exactly; with
105 $n \approx 10(m - 1)$ these instances defeat branch-and-bound at remarkably small
106 sizes. The mechanism is the integrality gap: the relaxation polytope is large
107 and its vertices are fractional, so bounding prunes little and the search must
108 branch nearly blindly. Aardal et al. [2] showed that lattice basis reduction
109 attacks the same family far more effectively than branch-and-bound, a result
110 we rely on in Section 3.2.

111 Our generator (Section 4.1) makes this mechanism explicit by maximizing
112 the spread of relaxation vertices, which enlarges $\tilde{\mathcal{S}}$ and weakens the relaxation
113 signal by construction.

114 2.2. Complete solvers

115 *Mixed-integer programming.* Branch-and-bound over the linear relaxation,
116 strengthened by presolve and cutting planes, is the standard approach. Its
117 leverage comes from pruning nodes whose relaxation is infeasible or bounded
118 away from the target, which is precisely what the market-split construction
119 removes.

120 *Constraint programming with clause learning.* CP-SAT-style solvers com-
121 bine constraint propagation with conflict-driven clause learning [12, 14] and
122 lazy clause generation [15]. Their strength on this family comes from a
123 property central to our argument: a search *manufactures* information as it
124 descends. A marginal $\Pr[x_j = 1 \mid A, \mathbf{b}]$ may be uninformative at the root
125 while $\Pr[x_j = 1 \mid A, \mathbf{b}, x_1=1, x_5=0]$ is 0 or 1. Structure invisible at the root
126 becomes deducible after commitments are made, which is why Section 3.4
127 studies the interior of the tree rather than the root.

128 *Lattice reduction..* An alternative attack embeds (1) in a lattice and searches
 129 for short vectors. LLL [10] reduces a basis in polynomial time and BKZ
 130 [16] generalizes it with a block size trading time against basis quality. The
 131 Aardal–Hurkens–Lenstra embedding [1, 2] is constructed so that a 0/1 so-
 132 lution appears directly as a short vector, making reduction a certificate-
 133 producing procedure rather than a heuristic. We use it as the first stage
 134 of the deployed pipeline. Lattice enumeration remains the state of the art
 135 on the canonical benchmark: Wassermann [35] solves QOBLIB market-split
 136 instances up to $m = 14$ on one CPU where branch-and-cut stops at $m = 7$;
 137 our instances are a different regime (multiplicity matched, $m/n \approx 0.35$) and
 138 our contribution is a branching methodology rather than a record there.

139 2.3. Learning for combinatorial optimization

140 Bengio et al. [4] survey this field and frame it in terms we adopt: solvers
 141 rely on hand-crafted heuristics for decisions that are “otherwise too expensive
 142 to compute or mathematically not well defined,” and learning is a candidate
 143 for making those decisions better. The phrase *too expensive to compute*
 144 presumes the decision is computable in principle and asks only about cost,
 145 which is the situation we measure in Section 4.5.

146 *Marginals of the solution set as a branching signal..* The fraction of solutions
 147 agreeing with a decision was a branching criterion before learning: counting-
 148 based search [32] branches on the variable–value pair of maximum solu-
 149 tion density computed inside individual constraints, and belief-propagation-
 150 guided decimation [29] fixes the most biased variable under approximate
 151 marginals recomputed after each fixing. Learned versions replace the count-
 152 ing by a network: NSNet [28] trains a BP-structured graph network on exact
 153 marginals of a SAT formula’s satisfying assignments, obtained by enumer-
 154 ation, and rounds them into a local-search initialization; MIP-GNN [27],
 155 Neural Diving [30] and predict-and-search [26] predict per-variable biases
 156 from collected near-optimal solutions and use them to fix variables, warm-
 157 start or restrict a search. Our target is the same as NSNet’s; what differs is
 158 that the marginals are conditional, carried to every node by the reduction
 159 identity, and used to guide a complete search without fixing anything, and
 160 that the DET/OPEN partition and the in-tree-versus-root comparison have no
 161 counterpart there.

162 *Branching imitation..* The clearest success in this area is imitation of an
 163 expensive expert. Strong branching scores candidates by tentatively solving

164 both children’s relaxations; it yields small trees and is too slow to run at
 165 every node. Gasse et al. [7] train a bipartite graph network to imitate it and
 166 recover most of the benefit at a fraction of the cost, following Khalil et al.
 167 [9]. The gain is not new information — the expert already knew the answer
 168 — but cheap access to it. Our method occupies the same structural position,
 169 with two differences: the target is an exact posterior rather than an expert’s
 170 scores, and the analysis identifies which of the expert’s decisions matter.

171 *Direct satisfiability prediction..* Attempts to have a network decide satisfia-
 172 bility end-to-end have a weaker record. NeuroSAT [17] classifies satisfiability
 173 by message passing over a literal–clause graph and succeeds on small ran-
 174 dom instances, but Li et al. [11] report substantial accuracy collapse under
 175 size transfer. Chen et al. [5] prove that standard message-passing networks
 176 cannot separate certain feasible/infeasible MILP pairs at all, since the pairs
 177 are indistinguishable in the Weisfeiler–Lehman hierarchy that bounds such
 178 architectures [36]. We therefore do not ask the network for a feasibility ver-
 179 dict; it supplies branching guidance inside a complete search that remains
 180 responsible for soundness.

181 *Graph representations..* We use graph attention [18] over the bipartite variable–
 182 constraint graph, the standard encoding for MILP in this literature [7].

183 *Reinforcement learning for branching..* Imitation targets an expert’s rank-
 184 ing, not the search cost the ranking is meant to reduce. FMSTS [20] uses
 185 a node’s subtree size as an observable Q-value, shows that under depth-
 186 first search minimizing every subtree minimizes the whole tree, and learns
 187 from scratch; tree MDPs [21] formalize that per-subtree credit assignment;
 188 retro branching [31] learns from retrospectively extracted subtree trajecto-
 189 ries; SORREL [25] initializes from suboptimal demonstrations; Neuro# [34]
 190 trains a graph network on the residual formula of a #SAT solver by evolution
 191 strategies to minimize branching decisions. In constraint programming, value
 192 ordering has been learned from solutions [24] and by deep Q-learning [23],
 193 and impact-based search [33] prefers the variable whose assignment shrinks
 194 the search space most. Our fine-tuning stage (Section 3.8) uses the same
 195 per-subtree credit with a plain policy-gradient update [19], and differs in its
 196 starting point: the policy is initialized at a supervised rule derived from exact
 197 conditional marginals, and the marginal network is frozen, so the only thing
 198 the cost signal can change is which variable is branched on. The classical

199 statement of the objective is the fail-first principle [22]: choose the variable
 200 whose wrong branch is refuted soonest.

201 2.4. Amortized optimization

202 The distinction organizing our results predates its use in combinatorial
 203 optimization. Gershman and Goodman [8] introduced *amortized inference*:
 204 when many related queries must be answered, solving each from scratch is
 205 wasteful and the question becomes how to reuse computation across queries.
 206 Amos [3] develops the same idea as *amortized optimization* — using learning
 207 to predict solutions by exploiting shared structure across similar instances
 208 — and surveys its appearance in variational inference, meta-learning and
 209 control. Millidge [13] states the trade-off in the form most relevant here:
 210 *direct* optimization spends computation on the instance at hand and pays
 211 that cost at every invocation, whereas *amortized* optimization converts the
 212 problem into supervised learning with cheap inference but requires a corpus
 213 of solved instances up front and is bounded by the approximator’s gener-
 214 alization; the productive arrangement is hybrid, with amortized predictions
 215 steering a direct optimizer.

216 Our system instantiates that hybrid precisely. The direct method is LP-
 217 probing, sound and costly. The amortized method is the network, cheap and
 218 unsound. The hybrid rule is that the network steers branching while probing
 219 and propagation retain exclusive authority over variable fixing (Section 3.7).

220 3. Proposed Method

221 3.1. Setting and notation

222 For a partial assignment fixing variables $F \subseteq \{1, \dots, n\}$ to values $\mathbf{v} \in$
 223 $\{0, 1\}^{|F|}$, write

$$\mathcal{S}(F, \mathbf{v}) = \{\mathbf{x} \in \mathcal{S} : \mathbf{x}_F = \mathbf{v}\} \quad (3)$$

224 for the solutions consistent with it, and let $K = \{1, \dots, n\} \setminus F$ index the free
 225 variables. The *conditional marginal* of a free variable $j \in K$ is

$$p_j(F, \mathbf{v}) = \frac{|\{\mathbf{x} \in \mathcal{S}(F, \mathbf{v}) : x_j = 1\}|}{|\mathcal{S}(F, \mathbf{v})|}. \quad (4)$$

226 *3.2. System overview*

227 The deployed solver is a cascade of a lattice stage and a guided complete
228 search:

$$\text{propagation} \rightarrow \text{LP infeasibility rule} \rightarrow \text{kernel pump} \rightarrow \text{AHL/BKZ} \rightarrow \text{guided backtracking search} \quad (5)$$

229 The first three stages are cheap and sound. AHL/BKZ attempts a certificate-
230 producing lattice reduction with a size-dependent block parameter; when
231 it succeeds it returns a verified solution outright. The learned component
232 appears only in the final stage, and only in its branching decisions. This
233 placement matters for the evaluation: because the lattice stage is independent
234 of the branching heuristic, every heuristic under comparison receives exactly
235 the same residual set of instances, so conditioning on that residual describes
236 the algorithm rather than selecting favourable cases.

237 *3.3. Conditioning is reduction*

238 **Proposition 1.** *Let A_F and A_K denote the column submatrices of A indexed*
239 *by F and K . Then*

$$\mathcal{S}(F, \mathbf{v}) \cong \{\mathbf{y} \in \{0, 1\}^{|K|} : A_K \mathbf{y} = \mathbf{b} - A_F \mathbf{v}\}, \quad (6)$$

240 *the bijection being $\mathbf{x} \mapsto \mathbf{x}_K$. Consequently the conditional marginals (4)*
241 *of the original instance are the ordinary marginals of the reduced instance*
242 *$(A_K, \mathbf{b} - A_F \mathbf{v})$.*

243 *Proof.* $A\mathbf{x} = A_F \mathbf{x}_F + A_K \mathbf{x}_K$. Fixing $\mathbf{x}_F = \mathbf{v}$ gives $A\mathbf{x} = \mathbf{b} \iff A_K \mathbf{x}_K =$
244 $\mathbf{b} - A_F \mathbf{v}$, and $\mathbf{x} \mapsto \mathbf{x}_K$ is a bijection between the two sets since $\mathbf{x}_F$ is deter-
245 mined. The marginal statement follows by counting. $\square$

246 The identity is elementary but it is what makes in-tree supervision practi-
247 cal. A depth- d search node is not a special object requiring its own encoding;
248 it is an instance of the same problem on $n - d$ variables. One network, one
249 feature extractor and one label generator therefore cover the whole tree, and
250 training data at depth d is produced by substitution rather than by instru-
251 menting a solver.

252 Two implementation details preserve correctness. Rows that become all-
253 zero after substitution carry no constraint and are removed; retaining them
254 causes linear-programming backends to reject the model. And the free bound
255 condition $0 \leq \mathbf{b} - A_F \mathbf{v} \leq A_K \mathbf{1}$ is checked before any relaxation is solved,
256 since it detects a large fraction of dead nodes at negligible cost.

257 3.4. Which predictions can cost anything

258 At a node with surviving solution set $\mathcal{S}(F, \mathbf{v}) \neq \emptyset$, partition the free
259 variables by their conditional marginal:

$$\text{DET} = \{j \in K : p_j \in \{0, 1\}\}, \quad \text{OPEN} = \{j \in K : 0 < p_j < 1\}. \quad (7)$$

260 The distinction is operational rather than statistical:

- 261 • For $j \in \text{OPEN}$, both children of a branch on x_j contain solutions. Either
262 choice keeps a solution reachable, so *no branching decision on a OPEN*
263 *variable can cost anything*.
- 264 • For $j \in \text{DET}$, one child is empty. Branching against the forced value
265 guarantees that the subtree is exhausted without a solution, costing a
266 full backtrack.

267 This gives the conditional marginal its operational reading. It is not a
268 “probability of being correct” about some designated solution, but the *frac-*
269 *tion of the surviving solution set retained* by an assignment; and the only
270 errors that matter are those on variables where that fraction is 0 or 1.

271 Two consequences follow, both of which the experiments confirm. First,
272 an aggregate accuracy over all free variables is a poor objective, because it
273 mixes decisions that can cost a backtrack with decisions that cannot; the
274 relevant quantity is accuracy on DET. Second, since $|\mathcal{S}(F, \mathbf{v})|$ shrinks as the
275 search descends, the DET share grows with depth. This alone does not say
276 *where* a learned predictor is useful, because a DET variable that the relaxation
277 already rounds correctly needs no predictor. The useful region is where DET
278 variables are plentiful *and* the relaxation is weak, and Section 4.9 shows that
279 this is governed by constraint density $m/|K|$ rather than by depth as such:
280 on small instances the two coincide, on larger ones they do not.

281 3.5. Exact conditional marginals as training targets

282 For instances small enough to enumerate, $\mathcal{S}$ is obtained exactly with a
283 constraint solver in all-solutions mode, and (4) is computed by counting.
284 Training minimizes the cross-entropy to the resulting Bernoulli targets,

$$\mathcal{L} = -\frac{1}{|K|} \sum_{j \in K} \left[p_j \log \hat{p}_j + (1 - p_j) \log(1 - \hat{p}_j) \right], \quad (8)$$

285 which is minimized exactly at $\hat{p}_j = p_j$. Note that the same loss with a planted
 286 solution in place of p_j — the conventional choice — is a noisy surrogate
 287 whenever $|\mathcal{S}| > 1$, since the planted solution is one draw from the posterior;
 288 on our instances a Bayes-optimal predictor agrees with it only about two
 289 thirds of the time.

290 *Enumeration correctness.* A subtlety here is easy to get wrong and silently
 291 corrupts every label derived from it. Constraint solvers report distinct sta-
 292 tuses for “search exhausted” and “time limit reached with solutions already
 293 collected.” Only the former certifies that $\mathcal{S}$ is complete; accepting the latter
 294 substitutes a truncated solution set, which biases every marginal computed
 295 from it. In our implementation only **OPTIMAL** and **INFEASIBLE** are accepted
 296 and timed-out instances are discarded rather than averaged in. A useful
 297 diagnostic is that mean enumeration time equal to the time limit indicates
 298 truncated instances being counted as complete.

299 *Sampling reachable states.* A depth- d training state is built by drawing d
 300 uniformly from $\{0, \dots, 12\}$, taking a prefix of an LP-confidence variable or-
 301 dering, and fixing those variables to the values of a randomly chosen *actual*
 302 *solution*. Fixing arbitrary values would produce states with $\mathcal{S}(F, \mathbf{v}) = \emptyset$; such
 303 states are dead nodes that propagation prunes immediately and on which the
 304 conditional marginal is undefined, so they are not states a branching heuris-
 305 tic is ever asked about. The depth range is a design choice we did not tune;
 306 on 10×25 instances it yields training states with 13–25 free variables, and
 307 Section 4.9 examines what happens when the deployed search queries states
 308 outside that range, which at 21×60 is most of them.

309 3.6. Network

310 An instance is encoded as a bipartite graph with n variable nodes, m
 311 constraint nodes, and an edge (j, i) whenever $a_{ij} = 1$. Let $\tilde{\mathbf{x}}$ be the relaxation
 312 solution of the current (reduced) instance and $r_i = \sum_j a_{ij}$ the row sums.
 313 Node features are

$$\text{variable } j : (\tilde{x}_j, \deg(j)/m, |\tilde{x}_j - \text{round}(\tilde{x}_j)|), \quad (9)$$

$$\text{constraint } i : (b_i/r_i, r_i/n, (b_i - A_i\tilde{\mathbf{x}})/r_i), \quad (10)$$

314 i.e. relaxation value, column degree and fractionality for variables; tight-
 315 ness, scaled row size and relaxation residual for constraints. All are size-
 316 normalized, which is what permits a single parameter set to apply at any
 317 (m, n) .

Both node types are embedded to $h = 64$ dimensions by a linear map and refined by $L = 4$ rounds of bidirectional message passing with graph attention [18]:

$$\mathbf{h}^{\text{con}} \leftarrow \text{LN}(\text{ELU}(\text{GAT}(\mathbf{h}^{\text{var}} \rightarrow \mathbf{h}^{\text{con}})) + \mathbf{h}^{\text{con}}), \quad (11)$$

$$\mathbf{h}^{\text{var}} \leftarrow \text{LN}(\text{ELU}(\text{GAT}(\mathbf{h}^{\text{con}} \rightarrow \mathbf{h}^{\text{var}})) + \mathbf{h}^{\text{var}}), \quad (12)$$

followed by a two-layer head emitting one logit per variable, $\hat{p}_j = \sigma(z_j)$. The model has 21,761 parameters.

The architecture is deliberately small and single-headed. Its only output is the quantity the objective (8) defines; there are no auxiliary value or feasibility heads, because the search draws soundness from its symbolic components and asks the network for nothing else.

3.7. Inference: guidance, not commitment

The network is consulted once per node. Branching selects

$$j^* = \arg \max_{j \in K} |\hat{p}_j - \tfrac{1}{2}|, \quad \text{first value} = \mathbb{K}[\hat{p}_{j^*} \geq \tfrac{1}{2}], \quad (13)$$

that is, the variable the network is most confident about, tried at its predicted value. The search remains complete and a wrong prediction costs backtracking only.

Variables are *fixed* only by sound deduction. Two mechanisms are available: constraint propagation, and LP-probing, which fixes x_j when assigning the opposite value makes even the relaxation infeasible. The division of labour follows directly from the measurements in Section 4.5: where a wrong answer costs time, the cheap unsound predictor is preferable; where a wrong answer costs correctness, the expensive sound procedure is required and cannot be replaced.

Algorithm 1 states the resulting search.

3.8. Cost-aware fine-tuning of the branching policy

The rule of Algorithm 1 branches on the variable the network is surest about. That maximizes the probability that the first child contains a solution; it says nothing about how much the search pays when the first child does not, nor about how much smaller the children are. Both quantities are properties of the *subtree*, and a search produces them as a by-product: the cost of a branching decision at node u is the number of nodes expanded below u until

Algorithm 1 Guided backtracking search for a BLS

Input: instance $(A, \mathbf{b})$, guidance model f_θ , time budget T

stack $\leftarrow [(A, \mathbf{b})]$

while stack $\neq \emptyset$ **and** elapsed $< T$:

$(A', \mathbf{b}') \leftarrow \text{pop}(\text{stack})$

if $\mathbf{b}' < \mathbf{0}$ **or** $\mathbf{b}' > A'\mathbf{1}$: **continue** // *free bound check*

 drop all-zero rows of A' ; **if** none remain: **return** any completion

 propagate; **if** contradiction: **continue**

if propagation determined a set D of variables:

 push reduce($A', \mathbf{b}', D$); **continue** // *Proposition 1*

if all variables assigned: **return** solution if verified, else **continue**

$\hat{\mathbf{p}} \leftarrow f_\theta(A', \mathbf{b}')$ // *one forward pass, all variables*

$j^* \leftarrow \arg \max_j |\hat{p}_j - 1/2|$; $v \leftarrow \mathbb{K}[\hat{p}_{j^*} \geq 1/2]$

 push reduce($A', \mathbf{b}', \{j^*=1-v\}$), then reduce($A', \mathbf{b}', \{j^*=v\}$)

return unresolved

347 a solution is found there or the subtree is exhausted, and depth-first search
348 returns it exactly when it backtracks out of u . We use this cost to fine-tune
349 which variable is branched on.

350 *Decision process..* A state is the reduced instance $(A', \mathbf{b}')$ after propagation;
351 an action is a free variable j ; the first value tried remains $\mathbb{K}[\hat{p}_j \geq 1/2]$ from
352 the marginal network, which is not modified. The return of the action at u
353 is $-\log c(u)$ with $c(u)$ the subtree cost. Because every decision has its own
354 subtree, credit is assigned per decision rather than per episode, which is what
355 made the sparse-reward formulations we tried earlier unlearnable.

356 *Policy and value..* On the frozen MarginalNet trunk we add two heads. The
357 policy logit is

$$g_j(s) = \log|\hat{p}_j - \tfrac{1}{2}| + h_\phi(\mathbf{h}_j), \quad (14)$$

358 where h_ϕ is a two-layer network on the variable embedding $\mathbf{h}_j$ whose last
359 layer is initialized to zero, so that $\pi(j | s) \propto \exp(g_j/\tau)$ starts as a softening
360 of the supervised rule and its argmax is exactly Algorithm 1. The value head
361 $V_\psi(s)$ reads a pooled graph embedding together with $\log |K|$ and $\log m$ and
362 predicts $\log c$.

363 *Update..* Each epoch samples training states, rolls the sampled policy ($\tau = 1$)
364 out to a solution, and records every decision with its exact cost. With ad-

365 vantage $\alpha(u) = V_\psi(s_u) - \log c(u)$, standardized within the epoch, the update
 366 is

$$\mathcal{L} = \sum_u \left[-\alpha(u) \log \pi(j_u | s_u) - \lambda \mathcal{H}(\pi(\cdot | s_u)) + (V_\psi(s_u) - \log c(u))^2 \right], \quad (15)$$

367 with entropy weight $\lambda = 0.01$. Only ϕ and ψ are trained; the trunk and the
 368 marginal head are fixed, so the marginal used for the first value and for the
 369 analysis of Section 3.4 is unchanged. The critic is pre-trained on decisions
 370 recorded under the supervised policy before the actor is updated, and its
 371 held-out rank correlation with the true cost is used as a gate: if the baseline
 372 cannot rank costs, the advantage is noise and fine-tuning is not attempted.
 373 Epoch selection uses the total subtree cost on held-out training-distribution
 374 states, never the evaluation instances. At deployment the policy is used
 375 greedily: $j^\star = \arg \max_j g_j(s)$.

376 3.9. Cost model

377 The comparison that motivates the design is per-node cost for a whole-
 378 instance verdict, and its structure is asymptotic rather than a constant factor.
 379 Let n_c be the number of free variables at a node. The network solves *one*
 380 relaxation to build its features and performs *one* forward pass, so its LP
 381 count is $O(1)$ in n_c . LP-probing solves one relaxation per candidate variable,
 382 $O(n_c)$. The ratio between them therefore grows linearly in instance size, and
 383 any single number quoted for it is meaningful only together with the n_c at
 384 which it was measured.

385 This also means the comparison is sensitive to how the LP backend is
 386 implemented, in a way that cuts against the learned method. A backend
 387 that rebuilds the model for every call inflates probing’s cost by its full $O(n_c)$
 388 factor while charging the network only once, exaggerating the advantage. A
 389 correctly engineered branch-and-bound keeps one relaxation and re-optimizes
 390 from the parent basis after a bound change, which is an order of magnitude
 391 cheaper per probe. Section 4.5 measures both methods against a warm-
 392 started backend for this reason, and reports the ratio as a function of n_c
 393 rather than as a single figure.

394 4. Experiments

395 4.1. Instances and control of solution multiplicity

396 Instances are generated by the procedure $\text{GENHARD}(m, n, K)$: draw a
 397 random 0/1 matrix A of density $1/2$; plant $K = 20$ candidate solutions
 398 $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(K)}$ with $\mathbf{b}^{(k)} = A\mathbf{x}^{(k)}$; retain the one maximizing *vertex spread*,
 399 the mean pairwise L_2 distance among relaxation vertices obtained from ran-
 400 dom objective directions. Maximizing vertex spread enlarges $\tilde{\mathcal{S}}$ directly and
 401 is the mechanism that makes market-split instances hard for relaxation-based
 402 methods (Section 2). Uniqueness of the planted solution is not enforced. Ev-
 403 ery instance is feasible by construction: $\mathcal{S} \neq \emptyset$ is guaranteed by the planting,
 404 and deciding infeasibility is outside the scope of this paper (Section 4.14).

405 *Why $|\mathcal{S}|$ must be controlled across sizes..* Solution multiplicity changes the
 406 *character* of the prediction task, not only its difficulty. When $|\mathcal{S}| = 1$ every
 407 free variable is DET and predicting well is tantamount to solving; when $|\mathcal{S}|$ is
 408 large most variables are OPEN and the task is to estimate a diffuse posterior.
 409 Comparing sizes without controlling $|\mathcal{S}|$ therefore confounds size with task
 410 identity. Since $|\mathcal{S}|$ falls as m rises at fixed n , we select m per size to match
 411 it.

Size	m	m/n	median $ \mathcal{S} $	root ceiling	conditional ceiling
10×25	10	0.40	23	70.4%	87.0%
18×50	18	0.36	19	71.3%	87.2%
21×60	21	0.35	10	—	—

Table 1: Sizes used, with solution multiplicity matched. Ceilings (the Bayes-optimal per-variable accuracies of Section 4.3) agree to within one point between the two training sizes, so size is the variable that changes.

412 *Limits of the control..* Matching $|\mathcal{S}|$ forces m/n to vary ($0.40 \rightarrow 0.35$); at
 413 fixed n the two cannot be held simultaneously and we prioritize $|\mathcal{S}|$ because
 414 it determines task identity. The control is also granular: at $n = 60$ the
 415 median $|\mathcal{S}|$ steps $42 \rightarrow 12 \rightarrow 2$ across $m = 20, 21, 22$, so $m = 21$ is the best
 416 available integer and the realized evaluation set has median $|\mathcal{S}| = 10$ with
 417 $27/30$ verified.

418 *Where enumeration stops..* At $n = 70$ no enumeration completed within 120 s
 419 at the m giving the target multiplicity (0/10). At $n = 100$ the obstruction
 420 is more basic: a constraint solver finds *zero* solutions within 20 s at $m \in$
 421 $\{28, 32, 36, 40\}$ despite a planted one. At that size the bottleneck is finding
 422 any solution rather than choosing branches well, so a branching comparison
 423 there would not be informative. We therefore evaluate at $n \leq 60$ and report
 424 this boundary as a limitation.

425 4.2. Implementation and protocol

426 *Software..* Relaxations used for training features and for the original search
 427 loop are solved with SCIP via PySCIPOpt; the warm-started relaxations
 428 of Sections 4.5 and 4.6 use Gurobi 13 (dual simplex, bounds changed in
 429 place), which requires a licence — HiGHS exposes an equivalent incremental
 430 interface and should serve as a free substitute, though we have not verified
 431 the timings with it. Enumeration and the CP-SAT baseline use OR-Tools;
 432 lattice reduction uses fpylll (LLL and BKZ). The network is implemented in
 433 PyTorch with PyTorch Geometric (GATConv). One property of the relaxation
 434 deserves note: it carries no objective, so its solution is an arbitrary vertex
 435 of the relaxation polytope and different LP codes return different vertices.
 436 Both branching arms consume the same vertex within an experiment, so
 437 comparisons are fair, but absolute node counts differ between the SCIP- and
 438 Gurobi-backed runs.

439 *Training..* Adam, learning rate 10^{-3} , 20 epochs over 10,000 in-tree states,
 440 batch size 1 (instances vary in size), gradient-norm clipping at 1.0, loss (8)
 441 with exact marginal targets. Training states are drawn at depths 0–12 uni-
 442 formly, 6 states per source instance, from a pool of 1,760 instances; evaluation
 443 uses 440 instances from a disjoint pool. Training takes approximately 20 min-
 444 utes on one CPU core. *No hyperparameter was tuned:* width, depth, learning
 445 rate, epoch count and the training depth range are the values first tried, and
 446 there is no validation split. Reported figures are therefore not contaminated
 447 by selection on the test set, but neither is any of these settings claimed to be
 448 optimal.

449 *Threading..* All timing-sensitive components run single-threaded (`torch.set_`
 450 `num_threads(1)`). On graphs of this size multi-threaded BLAS is counter-
 451 productive: we measure 11.3 ms per forward pass at 8 threads against 0.9 ms
 452 at one.

453 *Search evaluation..* Each arm is run on identical instances with a hard per-
 454 instance wall-clock budget enforced by subprocess termination, since lattice
 455 reduction executes inside a blocking native call that in-process timers can-
 456 not interrupt. Instances run at concurrency 6 on a 20-core machine. This
 457 low concurrency is deliberate: wall-clock-budgeted comparisons are sensitive
 458 to CPU contention, and at higher concurrency we have observed instances
 459 solvable in 14–15s standalone exceeding a 20s cap.

460 *Statistics..* Solve-rate differences between arms on the same instances are
 461 tested with McNemar’s exact test; per-instance speed differences with a two-
 462 sided sign test. We report both rather than a single aggregate because they
 463 can have different power. The main comparison at 21×60 is replicated over
 464 three independent seeds and we report the spread; smaller-size and transfer
 465 results are single-run and marked as such. Seeds are set explicitly: an earlier
 466 version derived them from `hash()` of the instance filename, which Python
 467 randomizes per process, so those runs had no recorded provenance.

468 *Evaluation sets..* Two 21×60 sets are used: the original 30 instances (gen-
 469 erator seeds 700000–700029) and a 100-instance set (seeds 700000–700099)
 470 that contains them, built after the 30-instance results had been reported; $|\mathcal{S}|$
 471 is verified for 92/100, median 11. All numbers on the 100-instance set are
 472 lattice-off, three seeds. The training pool for the experiments of Sections 4.10
 473 and 4.11 is a separate set of 500 instances at 21×60 (seeds 800000–800499,
 474 split 400/100) whose overlap with every evaluation set is zero by hash of
 475 $(A, \mathbf{b})$; the 100 held-out instances of that pool supply the states on which
 476 fine-tuning epochs are selected.

477 4.3. Prediction quality against the Bayes ceiling

478 Because $\mathcal{S}$ is enumerated exactly, the Bayes-optimal per-variable accuracy
 479 is computable:

$$\text{ceil} = \frac{1}{|K|} \sum_{j \in K} \max(p_j, 1 - p_j), \quad (16)$$

480 attained by answering the majority value of the surviving solutions. No
 481 deterministic predictor of $\mathbf{x}^*$ from $(A, \mathbf{b})$ can exceed it. Table 2 evaluates
 482 predictors against it on 400 held-out 10×25 instances with $\mathcal{S}$ fully enumer-
 483 ated.

484 The network comes within 4.1 points of the ceiling overall and 6.6 on the
 485 top-3 subset, and it recovers most of the gap between relaxation rounding

Predictor	all variables	top-3	L_1 to true marginal
Bayes ceiling	70.8%	93.2%	0.0000
Proposed network	66.7%	86.6%	0.1216
LP relaxation rounding	60.9%	65.8%	—

Table 2: Root-level prediction against the exact posterior on the 182 root states of the held-out 10×25 evaluation pool. The network is the same checkpoint used in every search experiment below. “top-3” restricts to the three variables the predictor is most confident about.

and the ceiling ($60.9 \rightarrow 66.7$ against a ceiling of 70.8). Two preliminary experiments on an earlier pool of 10×25 instances, reported for completeness but not from the checkpoint evaluated here, bear on design choices: a 2.11M-parameter network with four output heads (selection, assignment, value and feasibility) scored 60.7% on root states, statistically indistinguishable from relaxation rounding at 60.6%, so capacity is not the binding constraint and a single-purpose head is preferable; and training the single-head architecture on exact marginals rather than the planted solution improved L_1 to the posterior from 0.1261 to 0.0977 while leaving argmax accuracy unchanged, which matters because the analysis of Section 3.4 consumes the marginal rather than its argmax.

What the aggregate conceals.. Table 3 decomposes the same root variables by the partition of Section 3.4.

Root variables	count	share	ceiling on subset
DET (branching against it empties the subtree)	38	6.3%	100.0%
OPEN (either branch reaches a solution)	562	93.7%	67.2%
all variables	600	100%	69.3%

Table 3: Root-level decomposition, 24 instances of 10×25 with $\mathcal{S}$ enumerated. The aggregate is 94% composed of variables on which no branching decision can cost a backtrack.

At the root, 93.7% of variables are OPEN. The aggregate ceiling of 69.9% is therefore dominated by predictions that cannot affect the search, and on the 6.3% that can, the information is already complete (100%). Root-level accuracy is consequently a weak objective for a branching heuristic, and the

operative question is not whether the information exists but what detects it
— taken up next.

4.4. Depth, and where the residual lies

Using Proposition 1 we measure, at each depth along a prefix drawn from an actual solution, the conditional ceiling and the fraction of remaining variables that propagation forces (Table 4).

depth	surviving $ \mathcal{S} $	conditional ceiling	proposed network	LP rounding
0	25.3	70.8%	66.7%	60.9%
4	4.2	82.3%	72.8%	68.5%
7	1.8	91.5%	81.2%	77.8%
8	1.4	94.4%	84.3%	80.9%
12	1.1	98.8%	96.8%	95.5%
all depths	—	87.2%	80.0%	76.5%

Table 4: Conditional information against depth on the held-out in-tree states of the 10×25 evaluation pool ($n = 25$), LP-confidence variable ordering. Same checkpoint as Table 2.

The consequential movement is in composition rather than level: the DET share rises from 6.3% at the root to 87.9% at depth 8 (Table 6). At the root almost every branching decision is inconsequential because both children contain solutions; by depth 8 almost every decision empties a subtree when wrong. The network’s margin over relaxation rounding is 3–6 points throughout and does not vanish at depth, which is the first indication that what it captures is not simply depth-dependent; Section 4.9 makes this precise.

Does in-tree supervision matter, controlling for everything else? A comparison between a root-trained and an in-tree-trained network is only informative if nothing else differs. Table 5 holds the instance pool, the label type (exact marginals), the architecture, the optimizer and the number of training states (1,760 each) fixed, and varies only whether the states are root states or in-tree states drawn at depths 0–12. Both are evaluated on the same held-out in-tree states.

In-tree supervision is worth 4.4 points overall. The more telling row is the root-trained network itself: at 75.4% it is *below* relaxation rounding on in-tree states, so a network trained on root states does not merely underperform in the tree, it is worse than the classical heuristic it is meant to replace. The

depth	conditional ceiling	root-trained	in-tree-trained	LP rounding
0	70.7%	67.8%	66.5%	60.9%
4	82.1%	71.4%	72.4%	68.3%
6	89.7%	74.6%	78.2%	74.0%
8	94.4%	77.4%	84.2%	81.3%
10	97.1%	81.7%	91.4%	89.4%
12	98.8%	85.8%	96.5%	95.3%
all depths	87.2%	75.4%	79.8%	76.4%

Table 5: Controlled comparison of training-state distribution. Same pool, labels, architecture, optimizer and state count; only the depth at which states are sampled differs. Evaluated on held-out in-tree states of the 10×25 pool.

shift from root to interior states is not a distribution the root-trained network generalizes across, and in-tree training is what makes the learned guidance better than the baseline at all.

Table 6 decomposes the residual gap at these depths and asks what each method detects.

depth	% DET	network on DET	propagation	LP integrality	LP-probing	network on OPEN
5	61.9%	89.7%	1.2%	67.9%	51.6%	54.4%
6	74.2%	88.2%	1.6%	70.4%	68.0%	52.5%
7	79.9%	88.8%	3.7%	73.3%	72.8%	51.2%
8	87.9%	89.4%	5.3%	76.9%	79.8%	53.8%

Table 6: Residual decomposition on the same states as Table 4. Columns 4–6 are recall on the ground-truth DET set (LP-probing measured with the warm-started backend of Section 4.5); columns 3 and 7 are the network’s argmax accuracy on each subset.

The gap lies entirely on DET variables: at depth 7, $0.80 \times 11.2 \approx 9$ points, matching the measured 10.3-point shortfall against the conditional ceiling. Performance near 50% on OPEN variables is correct behaviour rather than a defect, since those variables are genuinely undetermined. The striking figure is that unit propagation detects only 1.2–5.3% of logically determined variables: the information is present and derivable but not reachable by the cheapest sound rule. LP-probing recovers 52–80% and LP integrality 68–77%.

540 4.5. Cost per node

541 Both methods are charged everything they must pay to reach a whole-
 542 instance verdict: the network one relaxation for its features plus one forward
 543 pass, probing one relaxation per free variable. Both use the same warm-
 544 started backend, a single model whose variable bounds are changed in place
 545 and re-optimized from the parent basis (Gurobi dual simplex), which is how a
 546 branch-and-bound implementation reuses its relaxation. Timings are single-
 547 threaded.

Instances	n	network	LP-probing	ratio	probing recall
10×25	25	3.22 ms	5.60 ms	$1.7\times$	—
18×50	50	2.33 ms	11.84 ms	$5.1\times$	—
21×60	60	2.57 ms	18.39 ms	$7.2\times$	—
40×100	100	4.02 ms	62.46 ms	$15.5\times$	—
60×150	150	4.45 ms	79.06 ms	$17.8\times$	—
<i>in-tree states of 10×25 instances, by depth:</i>					
depth 5	20	2.44 ms	4.02 ms	$1.6\times$	51.6%
depth 6	19	2.39 ms	3.98 ms	$1.7\times$	68.0%
depth 7	18	2.35 ms	3.43 ms	$1.5\times$	72.8%
depth 8	17	2.34 ms	3.13 ms	$1.3\times$	79.8%

Table 7: Per-node cost for a whole-instance verdict, warm-started LP backend for both methods, single-threaded. The network’s accuracy on DET variables at depths 5–8 is 85.9–92.7% against probing’s recall in the last column.

548 The ratio behaves as the cost model predicts: it is negligible at $n = 25$
 549 and reaches $17.8\times$ at $n = 150$, tracking the $O(n_c)$ versus $O(1)$ difference
 550 in relaxation count. Two consequences follow, and they point in opposite
 551 directions.

552 *The advantage is real but asymptotic, and small where we can search..* At the
 553 sizes where our search actually operates ($n = 25$ – 60) the ratio is 1.3 – $7.2\times$.
 554 At the depths where prediction matters most (5–8 of a 10×25 instance,
 555 $n_c = 17$ – 20) it is 1.3 – $1.7\times$. A practitioner choosing between the two at
 556 that scale should prefer probing, which is *sound*: where its test fires the
 557 conclusion is a proof, whereas the network never offers a guarantee. The
 558 learned alternative becomes compelling only at $n \gtrsim 100$, and Section 4.1
 559 shows that search itself fails there for every strategy we tried.

560 *The measurement is sensitive to baseline engineering.* An earlier version of
561 this comparison used a backend that rebuilt the LP model on every call
562 and, additionally, excluded the network’s own feature relaxation from its
563 timer. That combination reported 19–23 $\times$ at depths 5–8, where the correctly
564 measured value is 1.3–1.7 $\times$ — an overstatement of more than an order of
565 magnitude, arising entirely from how the two sides were instrumented rather
566 than from what they compute. We report the corrected figures and note
567 the general hazard: in any cost comparison against a classical procedure, the
568 classical side must be implemented as a practitioner would implement it, and
569 the learned side must be charged for its own preprocessing.

570 4.6. Search performance

571 We compare branching strategies under matched wall-clock budgets with
572 the lattice stage disabled, isolating branching quality. LP selects the variable
573 with the most integral relaxation value and tries that value first; MODEL uses
574 the proposed network. Both run in the same loop: one warm-started relax-
575 ation per node (Section 4.2), identical propagation, identical value ordering,
576 so the arms differ only in which variable they branch on and which value
577 they try first. Each configuration is run three times with different seeds; the
578 depth-first search itself is deterministic given the relaxation vertex, so seeds
579 change timing but not node counts.

Size (budget)	strategy	solve rate	median nodes	median time	nodes/s
10 $\times$ 25 (60 s)	LP	100 $\pm$ 0.0%	46	0.03 $\pm$ 0.00 s	1,767
	MODEL	100 $\pm$ 0.0%	24	0.07 $\pm$ 0.01 s	376
18 $\times$ 50 (300 s)	LP	100 $\pm$ 0.0%	4,224	2.43 $\pm$ 0.33 s	2,066
	MODEL	100 $\pm$ 0.0%	1,085	1.82 $\pm$ 0.37 s	731
21 $\times$ 60 (600 s)	LP	100 $\pm$ 0.0%	82,366	51.5 $\pm$ 2.0 s	1,627
	MODEL	100 $\pm$ 0.0%	8,657	14.6 $\pm$ 0.7 s	619

Table 8: Branching strategy comparison on the warm-started loop, lattice stage disabled, 30 instances per cell, mean $\pm$ standard deviation over three seeds of the per-seed median. Node counts are seed-invariant.

580 Three points. First, the learned guidance produces a smaller tree at every
581 size — 2.0 $\times$, 3.9 $\times$ and 9.5 $\times$ fewer nodes than LP — and the ratio grows with
582 size, as the analysis of Section 4.9 predicts: the states where the network
583 separates from relaxation rounding are the large, weakly constrained ones

584 near the top of a big tree, and 10×25 has few of them. Second, a smaller
 585 tree buys time only once the tree is large enough to pay for the forward pass.
 586 At 10×25 the MODEL arm is *slower* (0.34–0.43 $\times$, faster on at most 1/30
 587 instances), because a 24-node tree costs less than the network evaluations
 588 that shrink it; its per-node cost is 2.6–4.7 $\times$ that of LP at every size, which
 589 is the forward pass of Section 4.5. At 18×50 the median per-instance ratio
 590 is 1.30–1.37 $\times$ in the network’s favour but not significant (faster on 18/30 in
 591 every seed, sign test $p = 0.36$). Third, at 21×60 the advantage is clear and
 592 replicates: the median per-instance ratio is 2.57 $\times$, 3.03 $\times$ and 2.96 $\times$ across
 593 seeds, the network is faster on exactly 23/30 instances in each ($p = 0.005$),
 594 and both arms solve every instance within budget.

595 Two further strategies were measured on an earlier version of this loop
 596 that rebuilt the relaxation at every node. RANDOM branching solved 0/30
 597 at 20×50 within 300s, and Section 4.12 shows it closes none of the $21 \times$
 598 60 residual instances on the present loop either. LP-PROBE, which probes
 599 every free variable and soundly fixes the forced ones before branching as LP,
 600 produced by far the smallest trees (51 $\times$ fewer nodes than LP at 20×50)
 601 and was the slowest arm, processing 6 nodes/s against 376: node count and
 602 wall-clock rank the methods in opposite orders, and a tree-size convention
 603 would select the worst of them. We have not repeated LP-PROBE on the
 604 warm-started loop; Section 4.5 measures its per-node cost there directly.

605 *Replication on 100 instances..* Thirty instances leave the sign test with little
 606 power and the median with wide uncertainty, so the 21×60 comparison was
 607 repeated on the 100-instance set (Table 9). The direction and significance
 608 hold — MODEL is faster on 70–71/100 instances ($p < 0.001$) with 5.5 $\times$ fewer
 609 nodes — but the margins are smaller than on the original thirty: 2.0–2.3 $\times$
 610 in time against 2.6–3.0 $\times$, and 5.5 $\times$ in nodes against 9.5 $\times$. The original
 611 thirty are the first thirty seeds of the larger set, and on them the supervised
 612 policy happens to do unusually well; we treat the 100-instance figures as
 613 the estimate and the 30-instance ones as the favourable end of its sampling
 614 variation.

615 We also record what the rebuilt-relaxation loop reported for the 21×60
 616 comparison, because an earlier draft of this paper rested on it: 30/30 against
 617 25/30 solved, and a 3.29 $\times$ median speedup. Warm-starting the relaxation
 618 removed the solve-rate difference entirely — the five instances LP failed within
 619 600s it now closes in 65–270s — and reduced the speedup. The solve-rate
 620 advantage was an artifact of a slow backend that penalized the arm expanding

strategy	solve rate	median nodes	median time	nodes/s	vs. LP (per instance)
LP	100 ± 0.0%	79,609	41.2 ± 1.9 s	1,966	—
MODEL	100 ± 0.0%	13,386	16.5 ± 0.6 s	796	2.0–2.3×, 70–71/100, $p < 0.001$
MODEL+FT (Section 4.11)	100 ± 0.0%	5,495	7.6 ± 0.3 s	738	4.8–5.9×, 81–84/100, $p < 0.001$

Table 9: 21×60 , 100 instances, lattice off, 600 s budget, three seeds (mean ± s.d. of per-seed medians; node counts are seed-invariant). The last row is the cost-aware fine-tuned policy of Section 4.11.

621 more nodes, and we do not claim it.

622 4.7. Comparison against complete solvers

623 Table 8 compares branching strategies inside our own search loop, which
624 measures branching quality but says nothing about how the system stands
625 against an off-the-shelf solver. That comparison is reported here, on the same
626 instances, single-threaded, under the same budget.

Size	CP-SAT	SCIP	proposed (MODEL)	CP-SAT advantage
10×25	0.002 s	0.009 s	0.07 s	35×
18×50	0.046 s	0.385 s	1.82 s	40×
21×60	1.159 s	3.687 s	14.6 s	13×
21×60 , 100 inst.	0.792 s	2.590 s	16.5 s / 7.6 s (fine-tuned)	21× / 10×

Table 10: Median solve time; all methods solve every instance at every size. The proposed column is the MODEL arm of Table 8 (mean of per-seed medians); the last row adds the 100-instance set and the fine-tuned policy of Section 4.11. The proposed system does not beat a complete solver at any size tested, and is outperformed by SCIP, whose LP code it uses for training features.

627 The gap is large and we state it plainly: *the contribution of this paper is*
628 *not a solver that competes with CP-SAT*. It is worth asking where the gap
629 comes from, because the answer bears on what the branching heuristic is
630 actually doing. Table 11 decomposes it at 21×60 into tree size and per-node
631 throughput, using CP-SAT’s own branch counter.

632 The learned guidance produces a *smaller* search tree than CP-SAT —
633 2.2× fewer nodes despite CP-SAT’s clause learning — and loses entirely on
634 per-node cost. Where that cost goes can be measured. Before the loop was
635 warm-started it processed 254 nodes/s, spending 4.34 ms per node rebuilding
636 and solving the relaxation; warm-starting the same LP brings that to 0.08 ms
637 and raises throughput to 619 nodes/s, which is the loop reported throughout

	CP-SAT	proposed	ratio
nodes (median)	18,827 branches	8,657	ours 2.2× fewer
throughput	19,782 nodes/s	619 nodes/s	CP-SAT 32× faster
total time (median)	1.065 s	14.6 s	14×

Table 11: Decomposition of the 21×60 gap. CP-SAT additionally learns a median of 5,514 conflict clauses per instance.

638 Section 4.6. Of the 1.6 ms that remain per node, the network’s forward pass
639 is 1.88 ms at the root and less at deeper nodes where the reduced graph is
640 smaller, and propagation and bookkeeping are under 0.1 ms: the forward
641 pass now accounts for over 90% of per-node cost, and we have not reduced
642 it. We therefore do not claim that the gap is an implementation artifact.
643 What the node counts establish is narrower: the deficit is not in branching
644 quality. Closing it would require at minimum a compiled inference path and,
645 as the next paragraph notes, algorithmic components our search lacks.

646 Two honest caveats attach to this reading. Tree sizes are not perfectly
647 commensurable — CP-SAT’s “branches” include restarts and conflict-driven
648 jumps, and each of its nodes performs more propagation work than ours
649 — so the 2.2× should be read as an order-of-magnitude statement. And
650 a competitive implementation would need more than a faster forward pass:
651 clause learning and restarts are absent from our search, and the last two are
652 what let CP-SAT scale past the sizes where we stop.

653 4.8. Size transfer

654 Because the features are size-normalized and the parameters do not de-
655 pend on (m, n) , one network applies at any size. We test whether that
656 transfers in practice, with $|\mathcal{S}|$ matched as in Table 1.

657 At the 21×60 evaluation size the network trained only at 10×25 — a 2.4×
658 smaller problem — is not worse than the one trained at 18×50 ; if anything
659 it is better, expanding fewer nodes on 18/30 instances and finishing faster on
660 18–19/30 in each seed, with a median per-instance time ratio of 1.24–1.31×
661 in its favour. Neither difference is significant (sign test $p \geq 0.20$), and both
662 solve 30/30. At 18×50 the picture is the same: the 10×25 network is faster on
663 16–17/30 instances ($p \geq 0.59$) although its median time is slightly higher, the
664 two summaries disagreeing because the paired differences are small against
665 instance-to-instance variation. A previous version of this comparison, on

Training size	Evaluation size	solve rate	median time	median nodes
10×25	10×25	100%	0.07 ± 0.01 s	24
10×25	18×50	100%	1.82 ± 0.37 s	1,085
18×50	18×50	100%	1.48 ± 0.15 s	942
10×25	21×60	100%	14.6 ± 0.7 s	8,657
18×50	21×60	100%	23.7 ± 0.7 s	14,742

Table 12: Transfer on the warm-started loop, lattice stage disabled, 30 instances per row, three seeds. Rows 2–3 and 4–5 each hold the evaluation size fixed and vary only the training size.

666 the rebuilt-relaxation loop, gave 15/30 and a $0.92\times$ ratio; the direction is
667 not stable across loops and we read the result as no detectable transfer loss
668 rather than as a transfer gain. Root-level prediction agrees: the 10×25
669 network scores 67.8%, 69.5% and 69.4% at 10×25 , 20×50 and 40×100 ,
670 holding its margin over relaxation rounding (+11.0, +7.2, +4.7 points) at
671 every size.

672 *Transfer of both stages beyond 21×60 .* Table 13 adds sizes above the training
673 range of every learned component, with m chosen for the constraint density
674 of 21×60 ($m/n \approx 0.35$) because solution multiplicity cannot be verified
675 by enumeration there. Stage 1 keeps a node advantage over the LP rule at
676 24×70 ($3.8\times$) and 28×80 ($2.4\times$) but its time advantage vanishes at 24×70
677 (0.7 – $1.1\times$ on instances both solve), since a 260,000-node tree pays for as
678 many forward passes. The cost-fine-tuned policies (Section 4.11), trained at
679 21×60 and at 18×50 only, solve 30/30 at 24×70 in every seed, 2.3 – $3.3\times$
680 and 2.1 – $2.7\times$ faster than Stage 1, and at 28×80 solve 18 and 17 of 30 within
681 1,200s against 8 for Stage 1 and 4 for the LP rule, where SCIP solves all 30
682 in a median of 112s; that size is the upper edge of the useful window.

size	measure	CP-SAT	SCIP	LP	MODEL (25)	MODEL+FT (60)	MODEL+FT (50)
24×70 (1200s, 3 seeds)	solved / median time	30/30 / 16s	30/30 / 12s	23–25/30 / 537–644s	29/30 / 358–457s	30/30 / 125–185s	30/30 / 135–165s
	median nodes	—	—	980,743	260,114	90,054	95,818
28×80 (1200s, 1 seed)	solved / median time	23/30 / 270s	30/30 / 112s	4/30 / >1200s	8/30 / >1200s	18/30 / 720s	17/30 / 852s
	median nodes	—	—	1,929,041	803,535	470,606	560,077

Table 13: Sizes above every training range, lattice off, 30 instances each; solution multiplicity unverified, constraint density matched. Training sizes in parentheses.

683 This matters practically because label generation is the binding con-
684 straint. Exact marginals require enumeration, which fails beyond $n \approx 60$;

transfer means a network trained where labels are obtainable can be deployed where they are not.

4.9. Where the transferred network helps

Section 4.4 located the useful prediction regime at depths 6–8 of a 10×25 instance. Whether that localization survives transfer is a separate question, and it does not: at 21×60 the search queries the network on states with 5–60 free variables, only 19.7% of which fall inside the training range of 13–25, and 80.2% of which are larger than any training state. Table 14 evaluates the network and relaxation rounding against exact conditional marginals on 21×60 in-tree states, grouped by the number of free variables, on the subset of states that could be enumerated within 15 s.

free vars	in training range	$m/ K $	network on DET	LP on DET	gain	corr(network, LP)
20	yes	1.05	100.0%	100.0%	+0.0	1.000
25	yes	0.84	100.0%	100.0%	+0.0	0.998
30	no	0.70	95.7%	95.0%	+0.7	0.954
40	no	0.53	75.5%	75.2%	+0.4	0.747
50	no	0.42	70.2%	65.0%	+5.3	0.664
60	no	0.35	93.3%	88.9%	+4.4	0.658

Table 14: 21×60 in-tree states by number of free variables, 9–14 enumerable states per row. Gain is the network’s margin over relaxation rounding on DET variables; the last column is the Pearson correlation between the two predictors’ outputs.

The pattern is the reverse of what a depth-based account predicts. Inside the training range the network is indistinguishable from relaxation rounding — both are perfect and their outputs correlate at 1.000 — because at $m/|K| \geq 0.8$ the reduced instance is over-constrained and the relaxation is already integral on every DET variable. The network’s entire margin arises on states it never saw during training, at 50–60 free variables, where the relaxation is weakest ($m/|K| \approx 0.35$ – 0.42) and its outputs have decorrelated from the relaxation’s. In terms of Section 3.4, this is the region where DET variables are present but the relaxation does not find them.

Two conclusions follow. The first corrects Section 4.4: depth was a proxy for constraint density that happened to hold on 10×25 , where descending from the root moves $m/|K|$ only from 0.40 to 0.59, and does not hold on 21×60 , where the same descent moves it from 0.35 past 1.0. The useful regime is defined by $m/|K|$, and at deployment size it sits near the *top* of the

tree. The second is that the transfer result of Section 4.8 is stronger than a matched-multiplicity comparison alone shows: the network extrapolates to states outside its training range in both size and constraint density, and that extrapolation is where its contribution lives. It is also a limitation, since the training range was not chosen with this in mind (Section 4.14).

4.10. Training on deployment-sized states: a negative result

Section 4.9 located the network’s margin on states with 50–60 free variables, outside the 13–25 range of its training states, and Section 4.14 of an earlier draft listed aligning the two as an obvious untested improvement. We tested it, holding everything but the training-state distribution fixed.

States.. From a separate 21×60 pool (Section 4.2) we built in-tree states with 30, 35, $\dots$, 60 free variables by fixing a prefix of the LP-confidence ordering to the planted solution, and labelled them with exact conditional marginals under the enumeration rule of Section 3.5. Every state with at most 55 free variables enumerated within 90 s; at 60 (the root) 291/400 did, and the failures are the instances with large $|\mathcal{S}|$, so root labels are biased toward small solution sets. A property of these states matters for what follows: with the prefix fixed to a solution, the surviving solution set is almost always a singleton once 50 or fewer variables remain (median $|\mathcal{S}(F, \mathbf{v})| = 1$, DET share 97%), so their labels are nearly hard. Three networks were trained with the architecture, optimizer, epochs and loss of Section 4.2: M0, the network used everywhere above (10×25 in-tree states); M1, the 2,691 deployment-range states only; M2, half of each. A pre-diagnostic on 10×25 alone, training on depth ranges 0–6, 6–12 and 0–12 with equal state counts, gave in-tree accuracies of 79.3%, 79.9% and 79.9%: within one size, the depth range does not matter.

Prediction.. Re-running the analysis of Table 14 with the three networks scored on identical states, the gain over relaxation rounding on DET variables at 50 and 60 free variables is +5.4 and +9.7 points for M0, +6.1 and +10.7 for M1, +5.9 and +10.6 for M2. Training in the deployment regime buys about one point there.

Search.. It costs far more elsewhere. On the 100-instance set M1 expands a median of 21,337 nodes against 13,386 for M0 ($1.6\times$ more; fewer on only 38/100, $p = 0.021$) and is $1.7\text{--}1.8\times$ slower; on the original thirty it expands 20,746 against 8,657. M2 sits between (18,558, $p = 0.089$). A switch that

745 routes nodes with ≥ 50 free variables to M1 and the rest to M0 recovers
 746 nothing (17,252 nodes, 43/99, $p = 0.23$). The mechanism is visible in the
 747 prediction diagnostics: on small, dense interior states (10×25 at depth 6–
 748 8, the kind the search spends two thirds of its nodes on) the probability
 749 that M1’s most confident variable is DET and predicted correctly falls from
 750 94.4% to 77.6%, and the rank correlation between its confidence and the true
 751 margin from 0.36 to 0.12, while on the deployment-range states it improves
 752 (77.2 $\rightarrow$ 86.0%). Nodes are where the interior is, and a network that is
 753 slightly better near the top of the tree and much worse inside it makes the
 754 search slower. We conclude that the training-state size distribution is not the
 755 lever, and that the residual identified in Section 4.9 is not a distribution-shift
 756 artifact.

757 4.11. *Cost-aware fine-tuning*

758 If neither accuracy on DET variables nor the training distribution explains
 759 what the supervised guidance leaves on the table, the natural suspect is the
 760 objective: the rule branches on the variable it is surest about, and being sure
 761 is not the same as being cheap. Three measurements test this, and the third
 762 turns it into a method.

763 *R0: does the confident choice cost the least?* On 100 held-out states with
 764 45–50 free variables we forced the root decision to each of the five variables
 765 the network is most confident about, ran the supervised search below it,
 766 and recorded the exact subtree cost of each. The confidence rule’s choice
 767 is the cheapest of the five on 22% of states; the median ratio of its cost to
 768 the cheapest candidate is 1.29 and the mean 2.97; on 24% of states some
 769 candidate is at least twice as cheap; and choosing the cheapest candidate at
 770 the root alone would remove 55% of all nodes summed over these states. The
 771 LP rule’s root choice costs the same as the network’s on only 10% of states,
 772 so the two rules are not merely tied. There is a large residual, and it is about
 773 *which* variable to branch on.

774 *R1: can the cost be predicted?* The critic is the baseline of the policy update,
 775 so it must rank decisions by cost on states it has not seen. Recording every
 776 decision of the supervised search on the training pool gave 37,275 decisions
 777 (9,375 on held-out states) with their exact subtree costs, of which two thirds
 778 occur at 25–34 free variables. A value head on the frozen trunk reaches a
 779 held-out Spearman correlation of 0.80 with $\log c$; a baseline that knows only

780 the state’s size (a linear fit on $\log |K|$) reaches 0.70, and within size buckets
781 the critic’s advantage is larger (0.68 against 0.38 at 35–44 free variables,
782 0.57 against 0.37 at 45–55). Unfreezing the trunk adds 0.02. The gate of
783 Section 3.8 is passed with the frozen critic.

784 *R2: fine-tuning.* Six epochs of 400 rollouts each ($\tau = 1$, 90 s budget, about
785 21,000 decisions per epoch) from the 45–60 free-variable training states. On
786 the 150 held-out monitor states the total subtree cost under the greedy policy,
787 relative to the supervised rule, was 0.79, 0.49, 0.56, 0.46, 0.65 and 0.56 after
788 epochs 1–6; epoch 4 was selected. Policy entropy fell from 1.73 to 0.10 over
789 the same epochs, so the policy is near-deterministic by the selected epoch.

790 *Evaluation.* Table 9 reports the fine-tuned policy (MODEL+FT) beside the
791 supervised one on the 100-instance set: 5,495 nodes against 13,386 ($2.3\times$
792 fewer, fewer on 70/100, $p < 0.001$ in every seed) and a median of 7.6 s against
793 16.5 s ($2.0\text{--}2.2\times$, faster on 64–67/100, $p \leq 0.007$), at an unchanged per-node
794 cost (738 against 796 nodes/s). Against LP-guided branching this is $4.8\text{--}5.9\times$
795 in time and $14\times$ in nodes. On the original thirty instances the effect is small
796 (8,873 against 8,657 nodes; $1.2\text{--}1.5\times$ in time, 18–21/30, p between 0.04 and
797 0.36): the seventy added instances account for the gain ($2.5\times$ fewer nodes,
798 53/70), which is consistent with the thirty being a favourable sample for the
799 supervised rule. In the deployed cascade (Table 15) the fine-tuned policy
800 closes the lattice residual in 7.3–11.7 s against 12.2–17.7 s for the supervised
801 one.

802 *Replication and transfer of the fine-tuned policy.* A second fine-tuning run
803 with a different training seed (different rollout sampling and initialization
804 of the correction head; epoch 2 selected by the same monitor) gives the
805 same picture on the 100-instance set — 4,940 nodes ($2.4\times$ fewer than the
806 supervised policy, fewer on 68/100, $p < 0.001$) and $1.9\text{--}2.4\times$ faster (66–
807 68/100, $p \leq 0.002$) — and a larger one on the original thirty (4,040 nodes,
808 $1.9\times$ fewer on 19/30; $2.3\times$ faster, 20–21/30). The two runs select different
809 variables at many nodes and reach similar costs, which is what one expects
810 if the objective, not a particular solution, is what was learned. The policy
811 was fine-tuned on 21×60 states only, yet at 18×50 it expands 424 nodes
812 against the supervised policy’s 1,085 ($1.7\times$ fewer, 20/30, $p = 0.099$) and is
813 $1.8\text{--}2.2\times$ faster (22–24/30, $p \leq 0.016$): the cost-aware improvement transfers
814 downward in size as the supervised one does.

815 *What changed..* On 300 held-out states the fine-tuned policy branches on a
816 different root variable than the confidence rule at 95% of them. Its pick is
817 typically the 15th most confident variable ($|\hat{p} - \frac{1}{2}|$ of 0.26 against 0.43 for the
818 rule’s pick), and it is DET with a correct first value less often (57.5% against
819 82.5%). The policy has moved *away* from the variables it is surest about,
820 and the search got cheaper. The improvement is not attributable to the root
821 decision alone: forcing only the root to the fine-tuned pick and running the
822 supervised policy below it gives a median cost ratio of 0.93 (cheaper on 22/46
823 disagreement states), so the gain is a property of the sequence of decisions
824 rather than of any single one. We can therefore say what the fine-tuning
825 optimizes — subtree cost, with the probability of being right held as a soft
826 prior — but not yet name the feature it found; a cheap proxy, the amount of
827 propagation the first child triggers, does not differ between the two choices.

828 4.12. Deployment in the full pipeline

829 Table 8 isolates branching by disabling lattice reduction. In deployment
830 the lattice stage runs first and the search sees only its residual. Because
831 that stage is independent of the branching heuristic, every arm receives the
832 identical residual set.

Size	strategy	closed by lattice	residual solved by search	median time on residual	end-to-end
10×25	both	30/30	—	—	100%
18×50	LP / MODEL	24–28/30	11/11 / 11/11	0.5–5.0 s / 0.6–2.5 s	100% / 100%
21×60	RANDOM	10–16/30	0/50	> 600 s	$44.4 \pm 8.3\%$
21×60	LP	10–16/30	50/50	52.9–65.1 s	$100 \pm 0.0\%$
21×60	MODEL	10–16/30	50/50	14.2–20.2 s	$100 \pm 0.0\%$
21×60	MODEL+FT	10–16/30	50/50	7.3–11.7 s	$100 \pm 0.0\%$

Table 15: Cascade results on the warm-started loop, three seeds at every size, 600s budget at 21×60 . Lattice coverage varies across seeds (16, 14, 10 of 30 at 21×60) because the lattice stage draws random column permutations, so the residual handed to the search varies with it; all arms within a seed receive the identical residual.

833 Lattice coverage falls with size — 30/30, 24–28/30, 10–16/30 — and that
834 decline is what creates room for branching quality to matter. At 10×25
835 and 18×50 the residual is empty or closed within seconds by either sound
836 strategy. At 21×60 between ten and twenty instances reach the search
837 depending on the seed, and the residual is genuinely hard: random branching
838 closes none of the 50 residual instances pooled over seeds, so the end-to-
839 end rate of that arm is the lattice coverage alone. Both relaxation-based

and learned branching close all 50, and the difference between them is time. On the residual the learned guidance is faster on 11/14, 14/16 and 15/20 instances in the three seeds (median per-instance ratio $3.48\times$, $2.99\times$, $2.38\times$; sign test $p = 0.057$, 0.004 , 0.041), 40/50 pooled with a median ratio of $3.00\times$ ($p = 2.4 \times 10^{-5}$), expanding $6.9\text{--}9.5\times$ fewer nodes. The residual is where the branching heuristic operates in deployment, and the speedup there matches the isolated measurement of Table 8.

4.13. Does the graph structure contribute?

MarginalNet consumes six scalars per node and propagates them over the bipartite graph. With a feature set that small, a per-variable model ignoring the graph is a serious baseline: if it matches, the message passing is decoration. Table 16 compares models trained on identical targets, loss and splits, differing only in whether structure is used. The aggregated variant appends mean and max of the constraint-side features over each variable’s incident rows, which is the information one message-passing round would deliver.

Model	accuracy	L_1 to marginal	parameters
Linear, variable features only	78.1%	0.2530	4
MLP, variable features only	77.8%	0.2522	8,641
MLP + constraint aggregates	79.4%	0.2361	9,025
MarginalNet (bipartite graph)	84.1%	0.1514	21,761
Bayes ceiling	87.3%	0.0000	—

Table 16: Ablation on in-tree states of 10×25 instances, 2,000 training and 800 test states.

Message passing contributes: +4.7 points over the best structure-free model and a 36% reduction in L_1 , closing roughly 60% of the residual gap to the ceiling that the MLP leaves open. The improvement in L_1 is the more relevant of the two, since the analysis of Section 3.4 consumes the marginal rather than its argmax.

The same table bounds the claim, though. A four-parameter linear model on the relaxation value alone reaches 78.1%, within 9.2 points of the ceiling; most of the available signal is in that single feature, and the graph adds a useful but not dominant increment on top of it.

865 4.14. Limitations

866 *Label generation bounds the method, not only its evaluation..* Exact marginals
 867 require enumerating $\mathcal{S}$, which does not complete beyond $n \approx 60$ on this fam-
 868 ily. Transfer permits deployment at larger sizes but not training there; ap-
 869 proximating p_j by solution sampling is the natural extension and is untested.

870 *The useful size window is bounded on both sides..* Below it, lattice reduction
 871 closes everything and branching quality is irrelevant. Above it the search
 872 fails: at 28×80 the fine-tuned policies solve 18/30 within 1,200 s and the su-
 873 pervised rule 8/30, and at $n = 100$ a constraint solver finds no solution within
 874 20 s. Our demonstration lives at $n = 50$ –70; sizes above 60 are matched on
 875 constraint density only and 28×80 is a single seed.

876 *The advantage is in time and tree size, not in solve rate..* On a warm-started
 877 loop both sound strategies solve every instance we evaluate within budget;
 878 what the supervised guidance buys at 21×60 is a 2.0–2.3 $\times$ median speedup
 879 and 5.5 $\times$ fewer nodes on 100 instances (2.6–3.0 $\times$ and 9.5 $\times$ on the original
 880 thirty, a favourable sample), and the fine-tuned policy a further 2.0–2.2 $\times$
 881 and 2.3 $\times$, significant per instance in every seed. An earlier draft reported
 882 a 5–0 solve-rate advantage from a loop that rebuilt the relaxation at every
 883 node; that difference vanished with the faster backend and we do not claim it.
 884 Whether a solve-rate advantage exists at larger budgets or sizes is untested,
 885 and with 30 evaluation instances per size the power to detect one would be
 886 low in any case.

887 *Multiplicity control is approximate..* Median $|\mathcal{S}|$ is 23, 19 and 10 across the
 888 three sizes; at $n = 60$ multiplicity steps $42 \rightarrow 12 \rightarrow 2$ across consecutive inte-
 889 ger m , so finer matching is unavailable. Three of thirty evaluation instances
 890 have unverified $|\mathcal{S}|$ and are marked as such. Matching $|\mathcal{S}|$ also moves m/n
 891 from 0.40 to 0.35.

892 *Cost advantage depends on scale and on baseline engineering..* The amortiza-
 893 tion argument holds asymptotically but is worth little at the sizes where our
 894 search operates: 1.3–1.7 $\times$ at the depths where prediction matters, against a
 895 warm-started probing baseline that is additionally sound. It reaches 15–18 $\times$
 896 only at $n \geq 100$, where search fails for every strategy we tested. We therefore
 897 cannot demonstrate the regime in which the argument is strongest.

898 *The search implementation is not competitive..* At 21×60 CP-SAT is $9\text{--}10\times$
899 faster than the fine-tuned policy and $20\times$ faster than the supervised one on
900 the 100-instance set ($13\times$ on the original thirty). Our branching produces
901 fewer nodes, so the deficit is per-node throughput ($32\times$), now dominated by
902 the network’s forward pass; we have not shown that a competitive implemen-
903 tation is achievable: a compiled inference path, clause learning and restarts
904 are all absent, and the latter two are load-bearing for CP-SAT at larger sizes.

905 *Training and deployment distributions barely overlap, and aligning them did*
906 *not help..* The training range $\{0, \dots, 12\}$ produces states with 13–25 free
907 variables, whereas the 21×60 search queries states with 5–60 and spends
908 80% of its queries above 25. Section 4.10 shows that training on the deploy-
909 ment regime makes the search slower; we do not know whether a different
910 construction of large training states (prefixes from the search’s own trajec-
911 tory rather than from a solution, or states with several surviving solutions)
912 would behave differently.

913 *The fine-tuning result rests on one training pool and few training seeds..* The
914 policy was fine-tuned on states from one 21×60 pool; two training seeds agree
915 at 21×60 and the gain transfers to 18×50 , but no larger size was tested
916 and both seeds select their epoch on the same monitor. Its entropy collapses
917 within four epochs, epoch selection is by a held-out monitor rather than
918 by any principled stopping rule, and we cannot yet name the feature the
919 policy exploits (Section 4.11). Whether the gain transfers across sizes as the
920 supervised one does is untested.

921 *Hyperparameters are untuned and there is no validation split for the super-*
922 *vised network..* See Section 4.2. The configuration is defensible as a first
923 attempt and is not contaminated by the test set; the depth range, at least,
924 was subsequently shown not to matter within a size (Section 4.10). The fine-
925 tuning stage does select an epoch on held-out states, which are drawn from
926 the training pool and disjoint from every evaluation instance.

927 *Every instance is feasible; deciding infeasibility is out of scope..* The condi-
928 tional marginal is undefined on an empty solution set, so Stage 1’s supervision
929 requires $\mathcal{S} \neq \emptyset$, and every instance we train or evaluate on has a planted so-
930 lution. Finding a solution quickly and refuting an infeasible instance quickly
931 are different objectives, and the second is where clause learning dominates.
932 Stage 2’s signal, by contrast, is defined either way — the cost of a decision

933 is the size of its subtree whether that subtree contains a solution or is ex-
 934 hausted — so extending the method to infeasible instances is naturally a
 935 Stage-2 question, and it is untested.

936 *Single instance family.* All results concern one generator from one appli-
 937 cation. Whether the amortization argument transfers to other constraint
 938 families is untested.

939 5. Conclusion

940 We presented a learned branching heuristic for binary linear systems to-
 941 gether with an exact procedure for supervising it. Two elements carry the
 942 method. Conditioning a BLS on a partial assignment is identical to reducing
 943 it, so a single network trained on ordinary instances predicts at every node of
 944 the search tree without architectural change. And on instances small enough
 945 to enumerate, the exact conditional posterior both supplies the training tar-
 946 get and identifies which predictions can affect the search at all: only variables
 947 that every surviving solution agrees on can cost a backtrack, and their share
 948 rises from 6.3% at the root to 87.9% at depth 8.

949 Measured inside our search loop the heuristic does what it is designed
 950 to do. It is $2.0\text{--}2.3\times$ faster in median than relaxation-based branching at
 951 21×60 on 100 instances in each of three seeds, expands $5.5\times$ fewer nodes,
 952 and closes the residual of the deployed pipeline $3.0\times$ faster, where random
 953 branching closes nothing; with solution multiplicity controlled, a network
 954 trained at $2.4\times$ smaller scale performs indistinguishably from one trained at
 955 the evaluation size, which matters because exact supervision is available only
 956 at small sizes.

957 We are equally explicit about what the results do not establish. The sys-
 958 tem does not compete with a complete solver: CP-SAT is $9\text{--}40\times$ faster at
 959 every size we tested. Decomposing that gap is informative rather than excul-
 960 patory — our branching yields $2.2\times$ fewer nodes and loses on throughput by
 961 $32\times$, which places the deficit in a Python research implementation whose per-
 962 node cost is now the network’s forward pass, and which lacks clause learning
 963 and restarts — but we have not shown that closing it is achievable. Nor does
 964 the cost argument carry the weight we initially assigned it. Against a prop-
 965 erly warm-started probing baseline the advantage is $1.3\text{--}1.7\times$ at the depths
 966 where prediction matters on our instances, and probing is sound where the
 967 network is not; the ratio reaches $15\text{--}18\times$ only at $n \geq 100$, a regime in which

968 search fails for every strategy we tried. An earlier version of this compar-
969 ison, using a backend that rebuilt the LP for each probe and excluded the
970 network’s own feature solve from its timer, reported 19–23× at a scale where
971 the true figure is near unity, and we record the correction because the error
972 came entirely from instrumentation rather than from what the two methods
973 compute.

974 Two further results sharpen what the supervised target is. Moving the
975 training states to the deployment regime, where the network’s prediction
976 margin lives, does not help and in fact doubles the search tree, because the
977 search spends its nodes in the small interior states the moved network no
978 longer handles well; the training-state distribution is not the lever. Optimiz-
979 ing the cost the search pays is. With the exact subtree cost of every decision
980 read off the search, an actor-critic fine-tuning of the branching choice —
981 initialized at the supervised rule, with the marginal network frozen — ex-
982 pands 2.3× fewer nodes and runs 2.0–2.2× faster than the supervised policy
983 on 100 instances, and the policy it finds branches on variables the network
984 is *less* sure about. The probability that a branch is right and the cost of
985 the branch being wrong are different objectives, and the second is the one a
986 search minimizes.

987 What remains, and what we claim, is methodological. For planted-
988 solution instance families the exact conditional posterior is computable and is
989 a well-posed supervision signal for branching, where the conventional target
990 — a single planted solution — is not. States drawn from inside the search
991 tree are a better training distribution than root states, and the reduction
992 identity makes generating them a matter of substitution rather than solver
993 instrumentation. The resulting heuristic transfers across instance size when
994 solution multiplicity is held comparable, and the exact cost of its decisions,
995 available from the search, is a second well-posed training signal that im-
996 proves it. Whether this becomes practically useful depends on two things
997 we have not done: extending exact supervision past $n \approx 60$, most plausibly
998 by approximating the marginals without exhaustive enumeration, and imple-
999 menting the search well enough that a branching improvement is visible at
1000 the system level.

1001 Reproducibility

1002 Code is organized by component under `proposed_src/`, with shared mod-
1003 ules in `util/` and a `ROLES.txt` in each directory describing every file. Scripts

1004 are run from their own directory. Every search number in Tables 8, 12
 1005 and 15 comes from one chain, v23_ablation/run_revision2.sh, which runs
 1006 the warm-started loop (v23_warm_search.py) over all conditions and seeds;
 1007 v23_aggregate.py produces the tables from its output.

```

1008 # instances with matched |S| and exact marginals
1009 cd proposed_src/v15_bayes
1010 python3 v15_gen_by_m.py --m 10 --n 25 --count 2200 --K 20 \
1011     --out ../../runs/v22/sols_10x25.json
1012
1013 # attach solution sets, then split 80/20 (seed 0)
1014 cd ../v22_transfer
1015 python3 v22_add_solutions.py \
1016     --inp ../../runs/v22/sols_10x25.json \
1017     --out ../../runs/v22/full_10x25.json
1018 python3 v22_split.py # -> full_10x25_{train,test}
1019
1020 # in-tree training states (depths 0-12, 6 per instance)
1021 cd ../v17_conditional
1022 python3 v17_make_conditional_data.py \
1023     --sols ../../runs/v22/full_10x25_train.json \
1024     --max_depth 12 --per_instance 6 \
1025     --out ../../runs/v22/cond_10x25_train.json
1026
1027 # train (about 20 minutes, single CPU core)
1028 python3 v17_train_conditional.py \
1029     --train ../../runs/v22/cond_10x25_train.json \
1030     --test  ../../runs/v22/cond_10x25_test.json \
1031     --n_train 10000 --n_test 2000 --epochs 20 \
1032     --target soft --out ../../runs/model_n25
1033
1034 # every search table: warm-started loop, 3 seeds,
1035 # conditions A/B/R/C/D/Crand x {lattice off, on}
1036 cd ../v23_ablation
1037 bash run_revision2.sh          # -> runs/v23/, chain.log
1038 python3 v23_aggregate.py      # -> runs/v23/aggregate.json
1039
1040 # v24: range training (negative), 100-instance
1041 # set, R0/R1/R2 fine-tuning and evaluation
1042 cd ../v24_deploy_range

```

```

1043 bash run_v24.sh; bash run_v24_switch.sh
1044 bash run_v24_r1.sh; bash run_v24_r2.sh
1045 python3 v24_aggregate.py \
1046     --tags M0 M1 M2 SW PV

```

1047 All generators are seeded. The released checkpoints are 96 KB each. In-
1048 stance data and weights are regenerated from the above rather than dis-
1049 tributed, and the full evaluation cycle is scripted in `run_revision2.sh` under
1050 `v23_ablation/`.

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